APEX

AI-Powered Simulated Patient & Empathy Feedback

"A single conversation can change a patient's life — but are we training for it?"

PROBLEM

- **Communication training** is **resource-intensive** and hard to scale.
- **Standardized patients** are **expensive** and have limited availability
- **Instructor feedback** is **inconsistent** and often delayed

Students need safe, repeatable practice to build real communication skills.

SOLUTION

APEX delivers scalable, AI-driven communication training for medical students through realistic AI-simulated patient conversations and automated empathy feedback.

AI Simulated Patient Conversations
Realistic, dynamic patient personas powered by large language models.

Interactive Chat-Based Interface
Accessible from any device, anytime, anywhere.

Automated Empathy Feedback
Instant, objective analysis of communication quality.

WHY IT MATTERS

- Poor communication leads to lower patient trust, worse outcomes, and safety risks.
- 70% of malpractice cases stem from miscommunication
- Training is limited and expensive, leaving students unprepared for real-world patient interactions

SYSTEM FLOW

Student Input
Types or speaks to initiate a scenario.

→

Conversation Engine
Processes input and maintains dialogue.

→

Simulated Patient Response
LLM generates realistic, emotional replies.

→

Analysis Engine
Evaluates empathy, tone, and communication skills.

→

Feedback Dashboard
Delivers scores, insights, and suggestions.

KEY FEATURES

- Roleplay with diverse simulated patients
- Scenario-based training (e.g., breaking bad news, informed consent)
- Real-time empathy & communication scoring
- Session history & progress tracking
- Research data export (anonymized)

DEMO SCENARIO

1. **Choose a Scenario**
   Student selects a case, e.g., delivering bad news
2. **Talk with the Patient**
   Engage in a realistic chat; the patient responds emotionally.
3. **Practice & Reflect**
   Student navigates the conversation, showing empathy and clarity
4. **Get Instant Feedback**
   Student selects a case, e.g., delivering bad news

EXTENSIBLE PLUGIN ARCHITECTURE

APEX uses a modular architecture that allows researchers to plug in custom models for patient simulation, empathy scoring, and dialogue evaluation. This supports experimentation with new AI approaches for medical communication training.

RESEARCH APPLICATIONS

- Studying doctor–patient communication
- Evaluating empathy training methods
- Testing new LLM dialogue models
- Comparing feedback algorithms

TEAM

Members:

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APEX Capstone Project
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Visit us at:
https://apexsimulator.org/

TECHNICAL ARCHITECTURE

Frontend:
React Web & React Native Movile

Backend API:
FastAPI (Python)

Database:
PostgreSQL / Supabase

AI Integration:
LLM for simulation & feedback

Authentication:
JWT-based auth & user sessions